QGIS Breeding Sites Data Sync Workflow

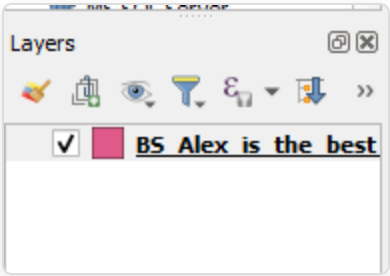
Overview

This workflow allows you to upload locally-edited breeding sites data (any `.gpkg`, or similar) directly into the production PostgreSQL database from QGIS, without manual field mapping or copy-paste errors.

Sync & Print Workflow

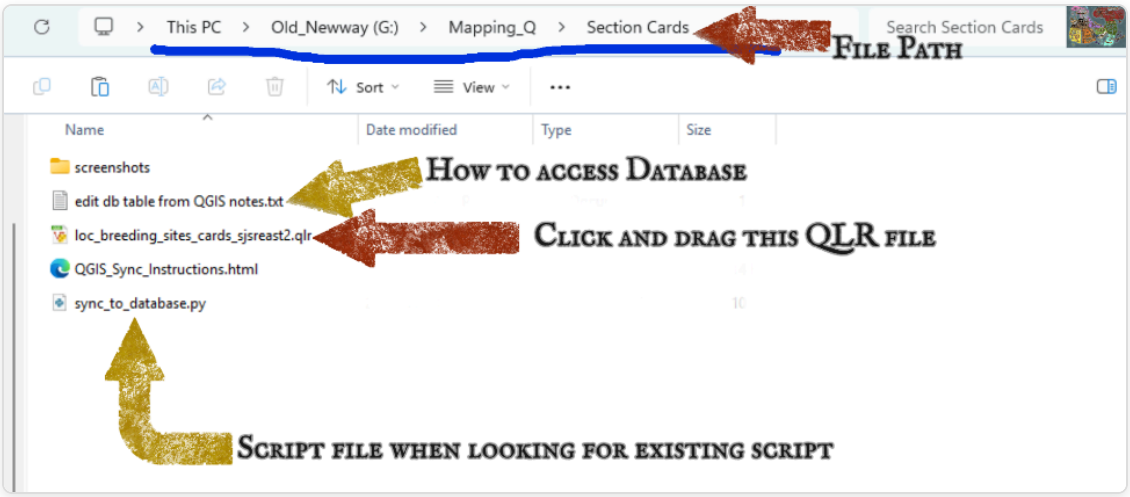
Step 1: Open QGIS and Load Your Data

1. **Open QGIS** (version 3.22 or later)
2. **Drag-and-drop your local data** into the map canvas
   - Example: Drag `BS_Alex_is_the_best.gpkg` from File Explorer into QGIS
   - Or: `Layer` → `Add Layer` → `Add Vector Layer` → Select your `.gpkg` / `.shp` file
3. **Verify the layer loaded**
   - You should see your features in the map
   - Layer name appears in **Layers Panel** (usually on left side)



Step 2: Load the Database Layer

1. **Locate the `.qlr` file** (see Prerequisites at the bottom of this page)
2. **Double-click the `.qlr` file** (or drag it into QGIS)
   - QGIS will prompt for database password
   - **Enter the mmcd_map password** (from `G:\Mapping_Q\Testing\Section Cards\`)
3. **Wait for connection to establish**
   - You should see a new layer: `loc_breeding_site_cards_sjsreast2` in the Layers panel
   - The map will center on the existing database features



Troubleshooting:

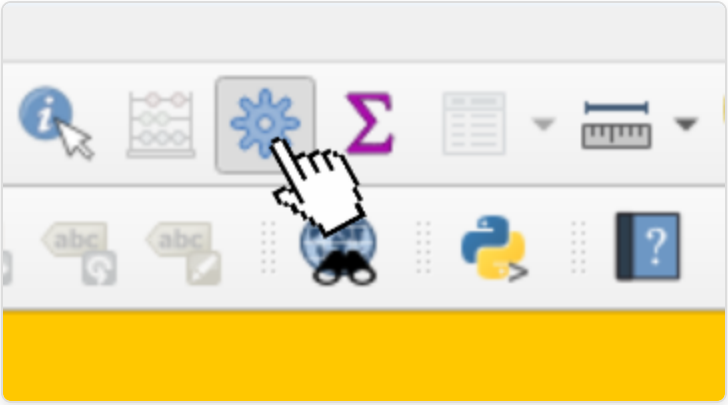
Issue	Solution
"Password rejected"	Verify password from `G:\Mapping_Q\Testing\Section Cards\` — copy exactly
"Connection timeout"	Check internet/VPN; database may be offline (contact admin)
"File not found"	Right-click `.qlr` → Open With → QGIS

Step 3: Open the Python Script in QGIS

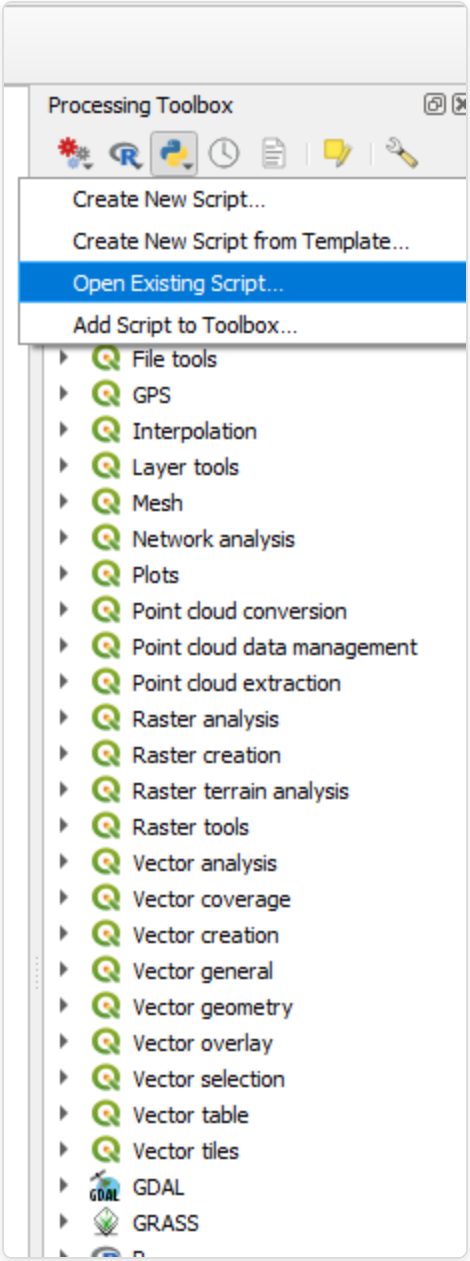
Using Processing Toolbox

1. Open **Processing Toolbox:**
   - Go to menu: `Processing` → `Toolbox`

- A panel opens on the right side

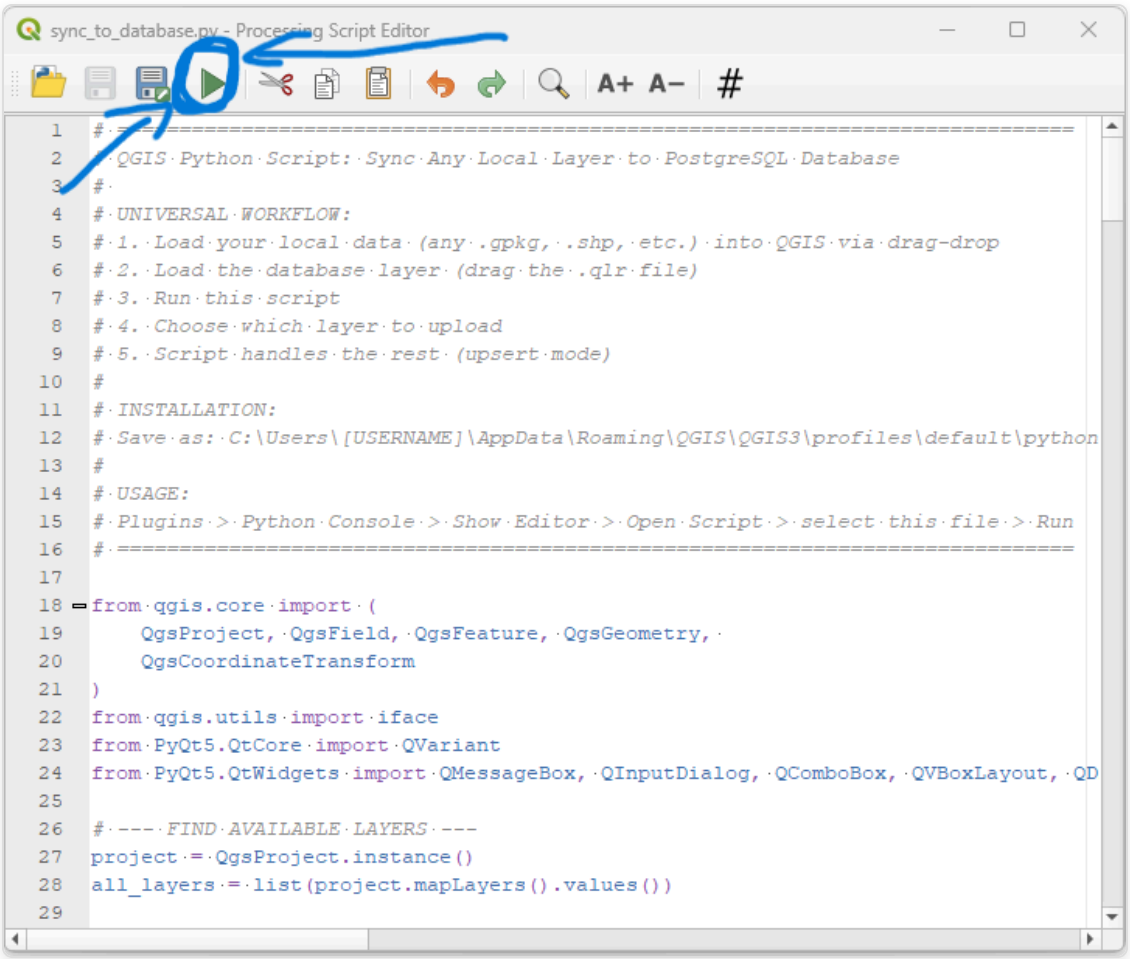


2. Find the Python button:
   - In the Processing Toolbox, look for the **Python** icon (snake symbol) or search box
   - Click the Python icon and select **Open Existing Script**
3. Navigate to the script:
   - Browse to: `G:\Mapping_Q\Testing\Section Cards\`
   - Select `sync_to_database.py`
   - Click **Open**



Step 4: Run the Script

1. **In the Python Editor, click the green Play button** (or `Run` → `Run Script`)
   - Script will execute and auto-detect your layers



2. **Choose your source layer** (if multiple local layers are loaded)
   - Dialog appears: "Which layer do you want to sync to the database?"
   - Keep the default if only one is loaded
   - Click **OK**
3. **Confirm the source layer:**
   - Dialog shows: "Found data layer: [name]. Use this as source?"
   - Verify the name matches your file (e.g., "BS_Alex_is_the_best")
   - Click **Yes**
4. **Confirm your selection:**
   - Script analyzes columns and reports what will be synced
   - Console output shows:

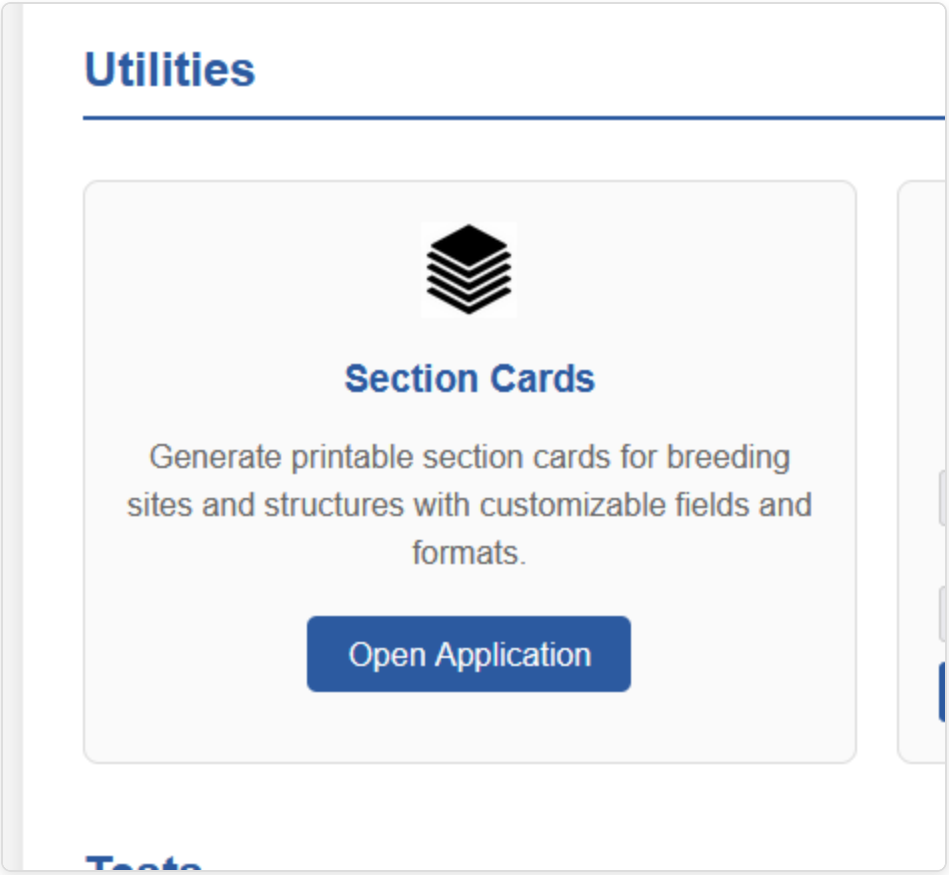
```
Source layer: BS_Alex_is_the_best (3099 features)
Target layer: loc_breeding_site_cards_sjsreast2 (3991 features)

Matching columns (53):
```

```
Sitecode -> sitecode
Foreman -> foreman
... (more columns)
```

Step 5: Open Section Cards App

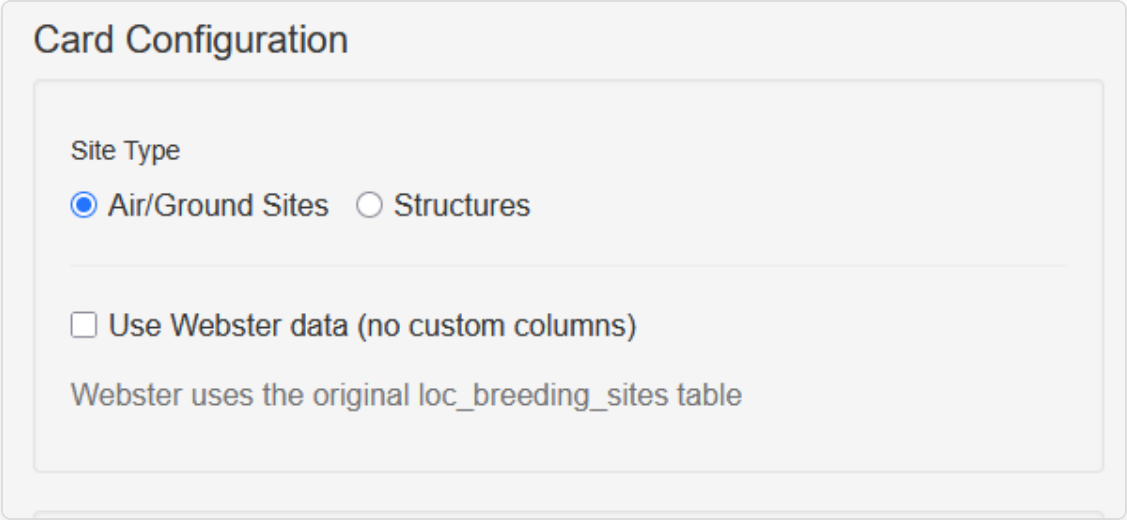
1. Go to **metrics.mmcd.org** in your browser
2. Scroll down to the **Utilities** section
3. You will see an app called **Section Cards** — click **Open Application**



Step 6: Choose Your Data Source

1. On the left sidebar, start by choosing your site type: **Air/Ground Sites** or **Structures**
2. If you uploaded your QGIS file to the database (Steps 1–4), keep **"Use Webster data"** unchecked — this lets you use your custom columns
3. If you did NOT upload via QGIS, check **"Use Webster data"** — you will only have what was in Webster as of March 22nd

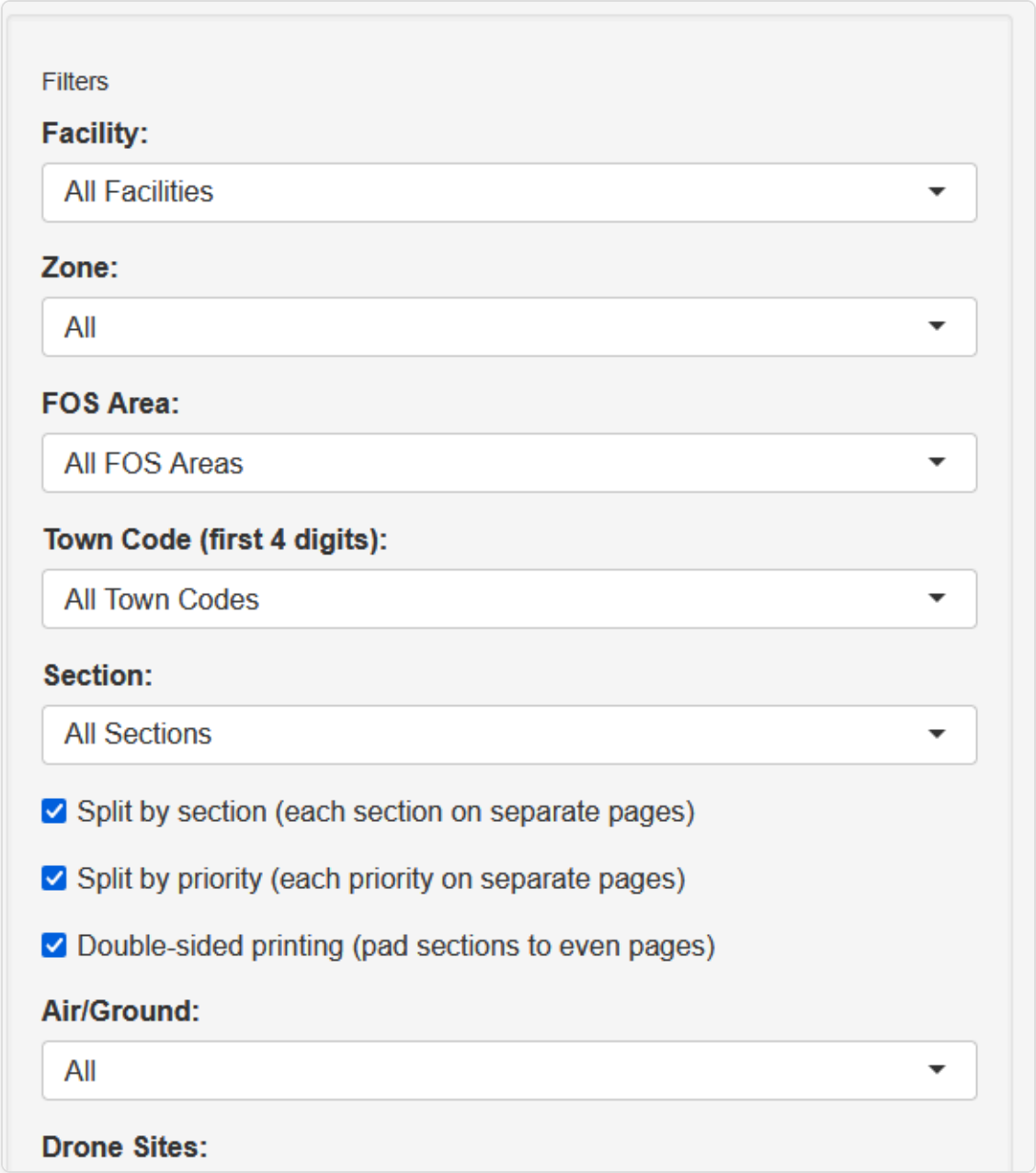
> **Note:** Custom fields are only available through Air/Ground sites. No custom fields yet for Structures.



Step 7: Set Your Filters

1. Pick your **Facility** and **FOS Area**
2. From there you can narrow it down further:
   - Only one **Town Code** or even just one **Section**
   - Only **Ground** or only **Air** or both
   - Include/exclude **Drone** sites
   - Filter by **Priority** color
3. **Split by section** and **Split by priority** are on by default — each section/priority group starts on its own page

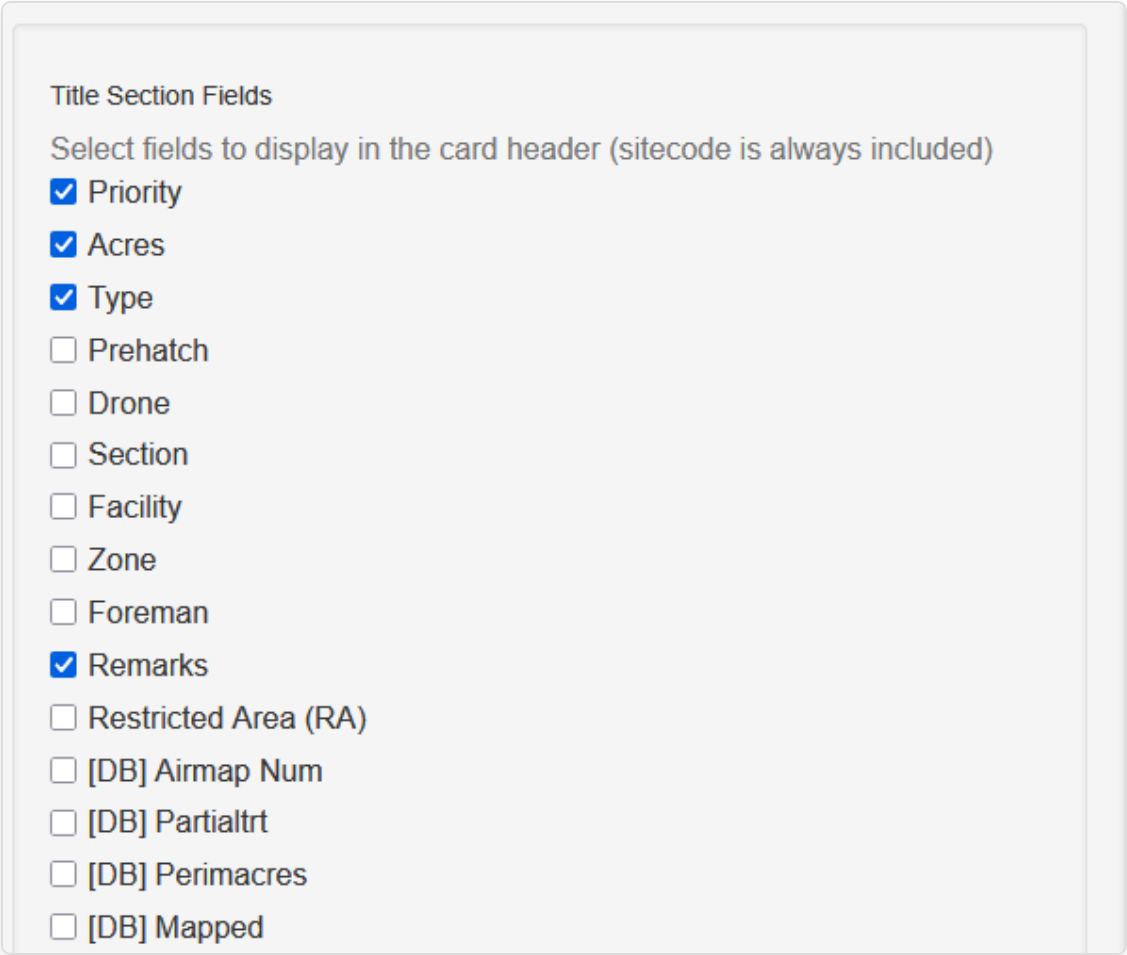
> **NOTE:** Most likely you want to print double-sided, so make sure **"Double-sided printing (pad sections to even pages)"** is checked (it's on by default).



Step 8: Pick Title Section Fields

These are the fields that appear in the card header — right next to the **Site Code**, above the data columns where inspectors fill out information.

- Examples: Priority, Acres, Type, Prehatch, Remarks, etc.



Step 9: Pick Watermark Fields

These appear **inside** the data table area as large, semi-transparent text overlaid on the columns.

- Culex, Spring Aedes, Perturbans, Sample Site, Prehatch Calculation
- They only show when a site's value is "Y" (or computed for prehatch)

Watermark Fields

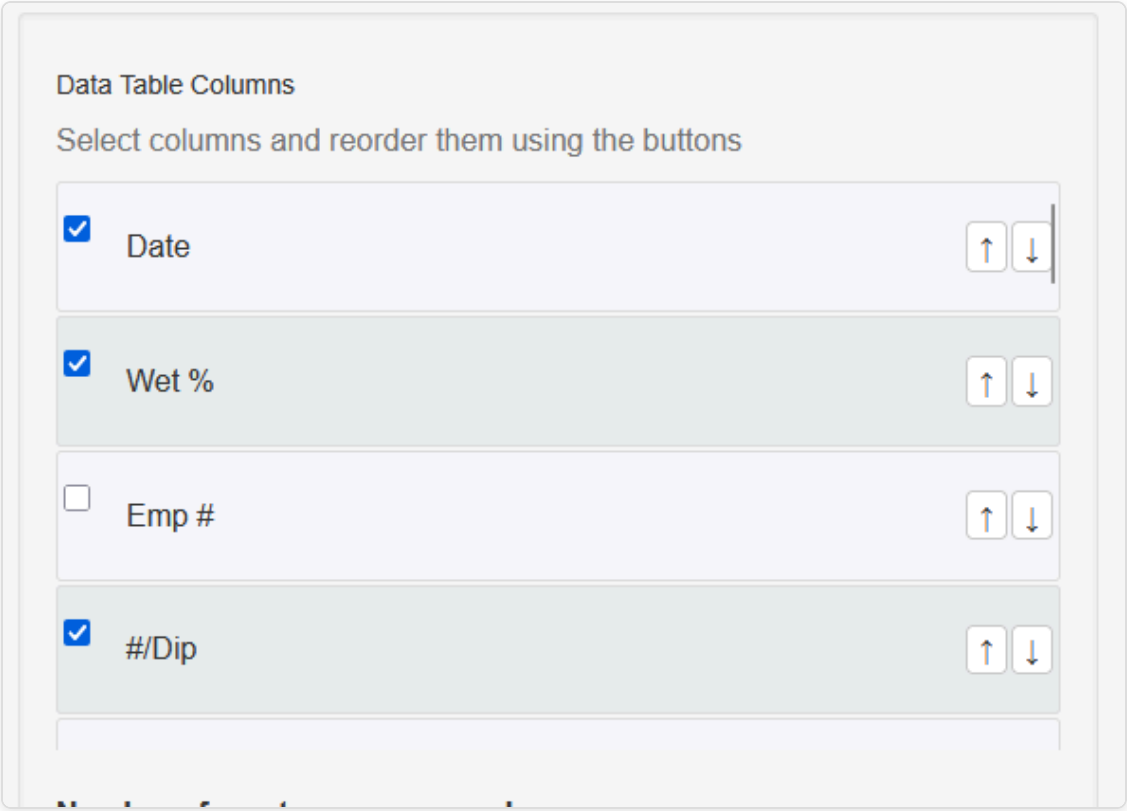
Semi-transparent labels overlaid at the bottom of each card (on top of columns)

- [ ] Culex
- [ ] Spring Aedes
- [ ] Perturbans
- [ ] Prehatch Calculation
- [ ] Sample Site

Step 10: Pick Data Table Columns

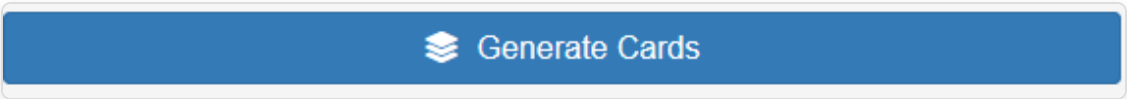
Choose which columns you want inspectors to **fill out by hand** on each card.

- If it's only for air dipping, you might turn off **Amt** and **Mat**
- If you created custom columns in your QGIS file, they'll show up here with a **[DB]** prefix — you can add them as table columns for inspectors to fill out
- Use the ↑ ↓ buttons to reorder columns however you like



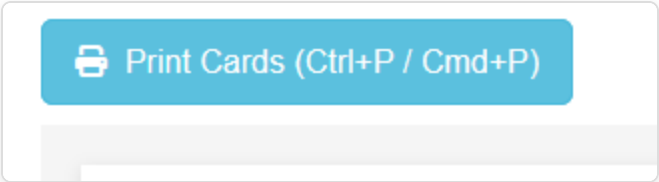
Step 11: Generate the Cards

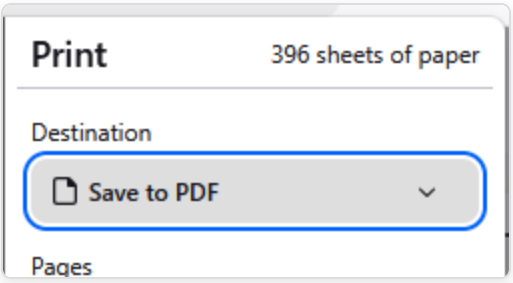
1. Click **Generate Cards**
2. Depending on the scope of what you're printing, this may take several seconds — be patient



Step 12: Preview & Print to PDF

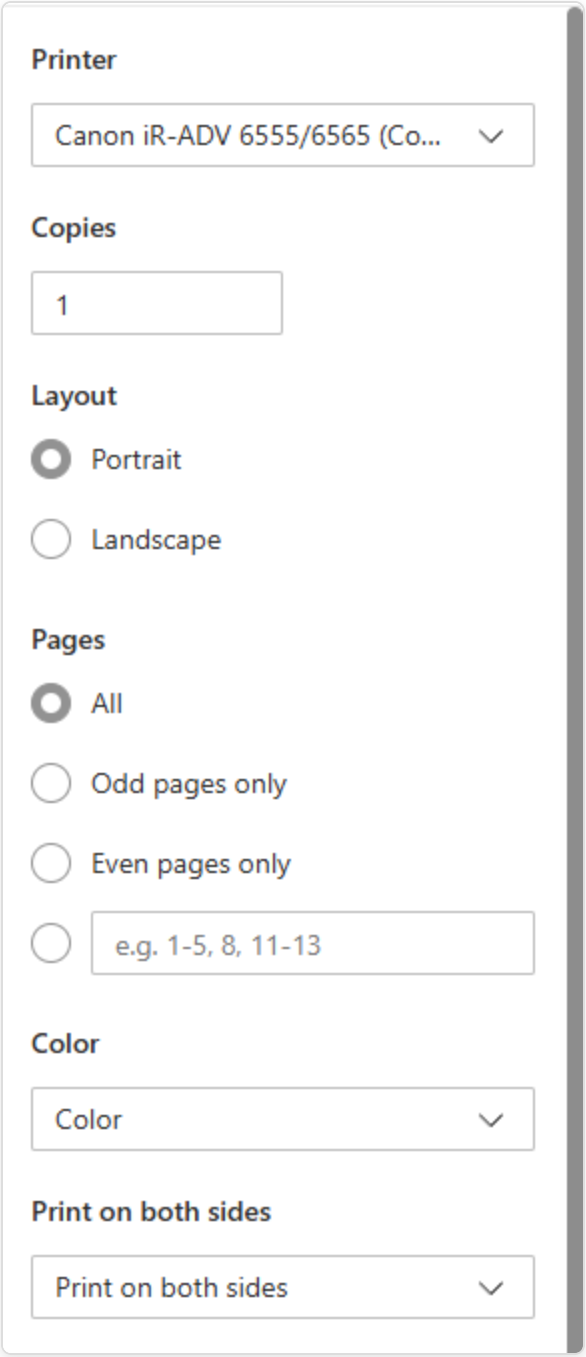
1. Once loaded, review the preview to make sure it looks right
2. Click **Print Cards** (or press Ctrl+P)
3. In the print dialog, choose **Save as PDF**
   - You might also be able to print directly from here — haven't fully tested it





Step 13: Print the PDF

1. The PDF may take **several minutes** to finish downloading, so be patient
2. Once it's done, open the file and print to whatever printer you choose
3. Make sure:
   - **Color** printing is on
   - **Double-sided** is turned on at the printer (if you want that)
   - **Headers and footers** are turned OFF in the print dialog



Step 14: Thank Alex Because He Is the Best

It says so in the screenshots, so it must be true.

Prerequisites & Setup

1. Locate Your Data Files

Your local breeding sites data should be in one of these locations:

Location	File Examples
Your local drive	`C:\Users\[YOUR_USERNAME]\Documents\BS_Alex_is_the_best.gpkg`
Shared network drive	`G:\Mapping_Q\Breeding Sites\BS_Alex_is_the_best.gpkg`
Project folder	`G:\[Project Name]\BS_Alex_is_the_best.shp`
Desktop	`C:\Users\[YOUR_USERNAME]\Desktop\BS_Alex_is_the_best.gpkg`

Note: The script auto-detects your file—no path configuration needed!

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2. Get Database Connection Info

Password location:

```
G:\Mapping_Q\Testing\Section Cards\
```

Look for a text file containing the `mmcd_map` password. **Keep this secure and do not share.**

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3. Get the Pre-configured QGIS Layer File

File location:

```
G:\Mapping_Q\Testing\Section Cards\loc_breeding_sites_cards_sjsreast2.qlr
```

What is it?

- `.qlr` file = QGIS Layer Definition (pre-configured with database connection)
- Contains write-access credentials for the database

Download to local QGIS folder (optional)

- If you prefer local copies, save to:

```
C:\Users\[YOUR_USERNAME]\AppData\Roaming\QGIS\QGIS3\profiles\default\python\scripts\sy
```

- Steps:
  1. Open File Explorer
  2. Navigate to `G:\Mapping_Q\Testing\Section Cards\`
  3. Copy `sync_to_database.py` to your local QGIS scripts folder (create if doesn't exist)